

Agora

AI Market Analyst

Framework: AutoGen + FastAPI + React | Version: 1.0 | Date: June 2026
Repository: bibhu2020/agenticaï (Hugging Face Space)

1. Overview

Agora is a two-phase AI market analysis platform with a React frontend and a FastAPI streaming backend. Given a stock ticker, Phase 1 deploys four independent data-collection agents in a round-robin team (Technical, Volatility, Sentiment, Fundamental analysts). Phase 2 sends their combined output to a strategy + risk decision team that uses a SelectorGroupChat to arrive at a final trade recommendation. Results stream to the browser in real time via Server-Sent Events (SSE).

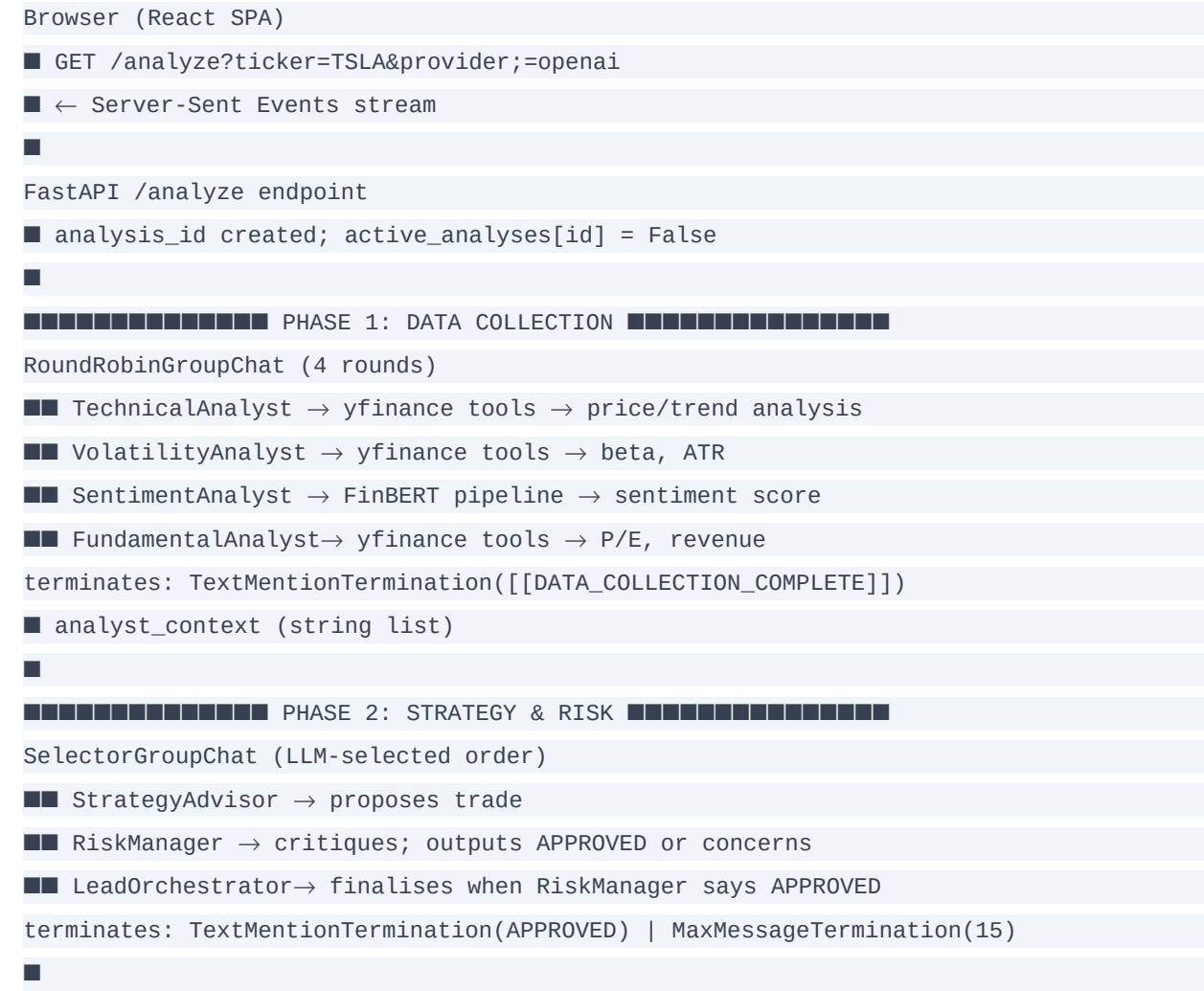
2. System Design

2.1 Components

Component	Technology	Purpose
frontend/	React + TypeScript + Vite	SPA: ticker input, real-time SSE message feed, structured result cards
backend/main.py	FastAPI + StreamingResponse	SSE endpoint /analyze; cancellation endpoint /cancel/{id}
backend/teams/team.py	AutoGen RoundRobin GroupChat / SelectorGroupChat	Phase 1 (analysts) and Phase 2 (decision) team factories
backend/aagents/market_analyst.py	AutoGen AssistantAgent	Technical analysis: price, MA, RSI, volume trends
backend/aagents/volatility_analyst.py	AutoGen AssistantAgent	Volatility metrics: beta, ATR, VIX correlation

Component	Technology	Purpose
backend/aagents/sentiment_analyst.py	AutoGen AssistantAgent	FinBERT-based sentiment + social signals
backend/aagents/fundamental_analyst.py	AutoGen AssistantAgent	P/E, EPS, revenue, debt ratios
backend/aagents/strategy_advisor.py	AutoGen AssistantAgent	Proposes or updates trade strategy
backend/aagents/risk_manager.py	AutoGen AssistantAgent	Critiques strategy; approves or blocks
backend/aagents/orchestrator.py	AutoGen AssistantAgent	LeadOrchestrator: terminates cycle when RiskManager approves
backend/tools/market_data.py	yfinance + Serper	Real-time price data and analyst reports
backend/tools/news_data.py	NewsAPI + FinBERT	News feed and NLP sentiment pipeline
backend/llm.py	AutoGen OpenAIChat CompletionClient	LLM factory — openai / google / groq

2.2 Architecture Diagram



```
SSE: data: [DONE]
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React: renders JSON structured result
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3. Agentic Flow

Phase	Team Type	Agents	Termination
1 — Data Collection	RoundRobinGroupChat	TechnicalAnalyst, VolatilityAnalyst, SentimentAnalyst, FundamentalAnalyst	[[DATA_COLLECTION_COMPLETE]] or 15 msgs
2 — Strategy & Risk	SelectorGroupChat (LLM-driven order)	StrategyAdvisor, RiskManager, LeadOrchestrator	APPROVED or 15 msgs

4. Streaming & Cancellation

- All agent messages are pushed to the browser in real time via SSE (media_type='text/event-stream')
- Each analysis session is assigned a UUID; POST /cancel/{id} sets a flag that breaks the async loop
- Structured JSON (risk assessment, final trade) is extracted and sent as 'structured_result' in SSE payload
- FinBERT sentiment pipeline is pre-warmed at FastAPI startup to reduce first-request latency

5. Configuration

Env Var	Required for
OPENAI_API_KEY	Default LLM (gpt-4o)
GOOGLE_API_KEY	Gemini 2.0 Flash option
GROQ_API_KEY	Llama 3.3 70B option
NEWS_API_KEY	News data tool
SERPER_API_KEY	Market data web search